

Model discrepancy formulation

Prior parameter distribution

$$\boldsymbol{\theta}_{\text{prior}} \sim \mathcal{N}(\mathbf{0}, \mathbf{W}_{\boldsymbol{\theta}}^{-1})$$

Discrepancy parameterization

$$\boldsymbol{\delta}(\mathbf{z}, \boldsymbol{\theta}) = \begin{bmatrix} \mathbf{I}_{n_u} & \mathbf{I}_{n_u} \otimes \mathbf{z}^{\top} \mathbf{M}_z \end{bmatrix} \boldsymbol{\theta}$$

Discrepancy noise model

$$\mathbf{s}(\mathbf{z}) - \mathbf{s}(\tilde{\mathbf{z}}) \sim \mathcal{N}(\mathbf{0}, \alpha \mathbf{M}_u^{-1})$$

Data and surrogate problem information

Large-scale / surrogate data

$$\mathbf{s}(\mathbf{z}_{\ell}), \tilde{\mathbf{s}}(\mathbf{z}_{\ell}), \quad \ell = 1, \dots, n_d$$

Solution to surrogate-constrained optimization problem

$$\tilde{\mathbf{s}}(\tilde{\mathbf{z}}), \tilde{\mathbf{z}}$$

Surrogate problem derivatives

$$\mathbf{v} \mapsto \tilde{\mathbf{s}}(\tilde{\mathbf{z}})^{\top} \mathbf{v}, \quad \mathbf{v} \mapsto \nabla_{\mathbf{z}, \mathbf{z}} \tilde{J}(\tilde{\mathbf{z}}) \mathbf{v}, \quad \dots$$

Discrepancy calibration
(Bayesian linear regression)

$$\boldsymbol{\delta}(\mathbf{z}; \boldsymbol{\theta}') \approx \mathbf{s}(\mathbf{z}) - \tilde{\mathbf{s}}(\mathbf{z})$$

Optimization solution update

$$\mathbf{z}' \approx \tilde{\mathbf{z}} - \mathbf{H}^{-1} \mathbf{B} \boldsymbol{\theta}'$$

Sensitivity-based update